

TPMS Explorer: an interactive, gate-verified open ecosystem for triply periodic minimal surface design, generative implicit fields, multifunctional physics simulation, and biophysical tissue-ingrowth dynamics

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Abstract

We present TPMS Explorer v7.0, a browser-based, zero-backend ecosystem for parametric design, physics simulation, and additive-manufacturing preparation of triply periodic minimal surface (TPMS) architected materials. The platform couples a watertight Surface Nets meshing pipeline with thirty-eight continuously executed CI gates plus a navigation quick-check (39-item suite, > 1000 assertions under strict per-line accounting, each gate carrying a minimum-assertion guard) claim shipped to the user. Version 7.0 adds five new grademodules: (i) a sinusoidal-activation implicit neural field (SIREN) with an exactly periodic Fourier-feature input and an 8-dimensional latent space over five distilled TPMS experts; (ii) multi-axial anisotropic yield envelopes (Hill-48, Tsai-Wu, Gurson, Drucker-Prager) with a unified star-convex ray-distance formulation and an interactive 3D viewer; (iii) a Bloch-Floquet phononic band-structure solver built on Born-von Krmn lattice dynamics with a realified two-pass de-plate-Lanczos eigen solution and long-wavedynamical-matrix calibration; (iv) a reaction-diffusion tissue-ingrowth engine coupling quasi-steady oxygen transport, hypoxia-gated osteoblast proliferation horizon; and (v) a level-set topology-optimization loop with Hamilton-Jacobi Godunov evolution driven by elastic and flow sensitivities and a side with zero dependencies beyond Three.js, export to industrial formats (STL, 3MF, GLB, Abaqus INP, OpenFOAM polyMesh, G-code), and are guarded by behavior-level audit gates whose calibration traps are documented. Post-release hardening adds (vi) a deterministic x-ray porosity solver---analytic-integration root finding with mesh-measured secant validation, reaching 0.26 pp deviation at resolution 96---and (vii) constructive global orientation propagation that reduces misoriented STL edges to zero by construction, replacing a paper-triangles normal voting scheme that degraded winding consistency five-fold.

Keywords: TPMS, implicit neural representation, SIREN, yield surface, phononic bandgap, tissue engineering, level-set topology optimization, continuous integration

1. Impact overview

Triply periodic minimal surfaces — first catalogued systematically by Schoen Schoen (1970) — are today a mainstay of architected-materials design and bone-scaffold engineering, where matching a target porosity while keeping the lattice manufacturable is the central numerical problem. TPMS Explorer turns a browser tab into a research bench for architected materials. The five v7.0 modules extend the platform from “geometry plus linear physics surrogates” to generative design, structural-failure envelopes, wave dynamics, biophysical growth, and closed-loop shape optimization — each guarded by an audit gate that executes the same code paths the user sees (Table 1).

2. Necessary resources

2.1. Hardware

Any 2020-class laptop. WebGPU is optional (acceleration only); all v7.0 modules run on the CPU path.

2.2. Software

Modern Chromium/Firefox/Safari; Node.js ≥ 20 for the CI suite; TypeScript 6, Vite 8, Three.js 0.185 (only runtime dependency).

3. Method design

3.1. SIREN implicit field with exact periodicity

A 4-layer sinusoidal network in the spirit of SIREN Sitzmann (2020), $F_\theta(\mathbf{x}) = W_3 \sin(W_2 \sin(W_1 \sin(W_0 \mathbf{u}))) + \mathbf{b}_3$ consumes the Fourier feature $\mathbf{u} = [\sin x, \cos x, \sin y, \cos y, \sin z, \cos z]$. Because every coordinate enters only through $\sin / \cos$, F_θ is exactly 2π -periodic (measured deviation 2.2×10^{-15}), which makes cross-period water-tightness a property of the architecture rather than of the fit. Five experts (Gyroid, Diamond, Lidenoid, Schwarz P, fractal trabecular) are distilled offline from the platform’s single semantic source of TPMS fields with Adam (RMSE 0.69%–8.04% at 16^3 , hardest for the multi-scale teacher). A latent code $\mathbf{z} \in \mathbb{R}^8$ selects experts through $m = \text{softmax}(-\|\mathbf{z} - \mathbf{A}_i\|^2 / 2\sigma^2)$ with ± 3 Walsh anchor codes (pairwise Hamming distance 4, cross weight e^{-72}). Lipschitz continuity is bounded by $\prod_i \|W_i\|_2$ (power iteration) and cross-checked against sampled gradient suprema.

3.2. Multi-axial yield envelopes

All four criteria are expressed in a unified *ray-distance* form: the distance $r(\hat{\mathbf{n}})$ from the origin to the envelope along a unit direction of principal-stress space. Hill-48 Hill (1948) (with

Table 1: v7.0 modules and their CI gates.

Stage	Module	Verified invariants	Assertions
I	SIREN implicit field	exact 2π periodicity (2.2×10^{-5}); central force network (measured ratio 0.53, matching the analytic 0.54). Under periodic boundary conditions rotations are chitz product bound vs. sampled gradients; latent-path Lipschitz; water-tight meshes at anchors and manifold midpoints	central force network (measured ratio 0.53, matching the analytic 0.54). Under periodic boundary conditions rotations are not admissible Bloch modes, so the Γ point carries exactly three translational zero modes.
II	Yield envelopes	uniaxial analytic anchors; midpoint convexity ($\times 200$ per criterion); isotropic permutation symmetry; Gurson hydrostatic pole; closed mesh volume by divergence theorem; safety factor of a half-scaled stress = 2	<i>3.4. Tissue ingrowth reaction–diffusion</i> Oxygen obeys a quasi-steady diffusion–Michaelis–Menten balance solved by 7-point Jacobi iteration with no-flux mirrors at solid interfaces and Dirichlet C_0 on the perfused boundary layer; osteoblasts follow hypoxia-gated logistic growth with explicit daily updates; mineral accumulates where oxygen exceeds a threshold. Operator splitting is justified by $\tau_{O_2} = L^2/D \approx 0.05$ days versus a 28-day proliferation horizon. $V_{\max}$ is calibrated to a literature oxygen penetration depth $L_{\text{pen}} = \sqrt{2DC_0/V_{\max}} \approx 0.1$ mm. The gate verifies the discrete mass balance (boundary-layer flux = interior consumption, residual 0.000%) and hypoxia gating on a dense-versus-porous pair.
III	Phononic bands	Γ -point triple translational zero modes; sound-speed self-consistency (gyroid 2.2%); per- k branch ordering; time-reversal symmetry $\omega(\mathbf{k}) = \omega(-\mathbf{k})$; diatomic-chain analytic gap; island-count guard	
IV	Tissue ingrowth	mass balance = boundary-layer flux – interior consumption (0.000%); core-oxygen transport differentiation; $\rho \leq \rho_{\max}$; hypoxia gating; mineral monotonicity; bitwise determinism	<i>3.5. Level-set topology optimization</i> The TPMS signed field evolves under $\partial\Phi/\partial t + V_n \nabla\Phi = 0$ (Godunov upwinding) with V_n assembled from elastic von-Mises sensitivities (fully-stressed) and Darcy pressure-gradient sensitivities, zero-measured over <i>interface</i> voxels, normalized, and constrained by a volume Lagrange bias. Classical compliance under a unit service load is proportional to 1/reaction of the displacement-controlled solve — the $\frac{1}{2}F\delta$ reading is a softening artefact under displacement control. Islands are removed by largest-component BFS, Φ is reinitialized by an exact 3D Euclidean distance transform, and the final field re-enters the platform’s Surface Nets pipeline through the precomputed-field <i>injection contract</i> , guaranteeing water-tight extraction.
V	Level-set optimization	unit-load compliance $\propto 1/\text{reaction}$; monotone under 0.5% tolerance; volume Lagrange constraint $\pm 10\%$; permeability gain $\times 1.50$; water-tight extraction (open edges = 0); reinitialization $ \nabla\Phi \approx 1$	

a hydrostatic cap), capped Drucker–Prager, and Gurson close to star-convex bodies containing the origin, hence r is single-valued and the safety factor is exactly $r(\hat{\mathbf{n}})/\|\sigma_0\|$. Tsai–Wu admits a closed-form quadratic root. Gurson’s hydrostatic pole Gurson (1977) $(2\sigma_0/3) \operatorname{arccosh}((1+q^2)/2q)$ with q equal to the porosity provides a strong analytic anchor. Scaffold-level parameters are derived from the same Gibson–Ashby constants Gibson (1997) used by the platform’s energy-absorption module.

3.3. Bloch–Floquet phononic bands

The unit cell is discretized into voxel masses connected by 26-neighbourhood axial springs; the Bloch–Floquet treatment follows the standard phononic-crystal formalism Hussein (2014). Bloch phases assemble a complex Hermitian $K(\mathbf{k})$, realified as $A - B$

BA so that all eigensolution machinery is real. Two-pass *deflate-Lanczos* (full reorthogonalization; the second run is deflated against the first run’s Ritz vectors) recovers both copies of every complex eigenvalue, preserving degeneracies. Spring stiff-

ness κ is calibrated by exact inversion of the long-wave dynamical matrix $\omega^2 = \kappa h^2 k^2 S_x / (mN^3)$, $S_x = \sum dx^4/d^2$ — affine homogenization is explicitly *not* the long-wave modulus of a

central force network (measured ratio 0.53, matching the analytic 0.54). Under periodic boundary conditions rotations are not admissible Bloch modes, so the Γ point carries exactly three translational zero modes.

3.4. Tissue ingrowth reaction–diffusion

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3.6. Exact porosity solver: analytic-integration root finding

The default porosity path couples the requested value to a voxel-rank bisection whose caliber differs from the delivered triangle mesh; measured deviations reached 24 pp for higher-harmonic families at coarse resolutions. Following the analytic-integration methodology of RegionTPMS Karakoç (2021), the agent CLI now (i) locates *iso** on the *analytic* field by bisection over a deterministic Monte-Carlo integration of $P(F < iso)$ in the unit cell (LCG-seeded, bit-reproducible), then (ii) builds once with *iso** (bypassing the internal bisection) and applies a single secant correction along the analytic slope to absorb the mesh-versus-analytic volume gap, reverting to the direct build if the corrected attempt is worse. Measured deviations: 0.26 pp (Diamond, 65% target, resolution 96) and 1.3–4.3 pp across cubic-symmetric families at resolution 48; higher-harmonic families at resolution 48 remain representation-limited (9–23 pp) and are reported honestly with an upgrade hint. A cube + Diamond + resolution-64 combination with non-manifold edges (nm = 740, down from 1940 under the legacy path) is fail-closed by the water-tightness gate.

3.7. Global orientation propagation

The binary and multi-solid STL writers previously flipped each triangle independently by comparing geometric normals against the vertex normal field; because that field is unreliable at cap fans and boundary skins, independent voting fragmented a locally consistent winding (misoriented edges $4536 \rightarrow 23571$ after flipping at resolution 96). The writers now call a constructive orientation pass: register every undirected edge with its two incident triangles, two-color the triangle adjacency graph by BFS (same-direction neighbours get opposite colors), apply the parity, and flip the whole mesh once if the divergence-theorem volume is negative. For the closed manifolds the pipeline guarantees (open = non-manifold = degenerate = 0), misoriented edges are zero by construction — verified byte-level on the exported STL (823,500 edges at resolution 96).

4. Quality control

Thirty-eight gates plus a navigation quick-check (39-item top-level suite) run via `npm run test:all` on three platforms. Every gate loads the shipped TypeScript through a bundler and asserts behavior-level invariants (no mocked math), and every gate now carries a *minimum-assertion guard*: a probe experiment showed that an auditor whose assertions are all neutralized still exits green under the historical `if (fail) exit` idiom, so each gate additionally fails when its executed-assertion count drops below a measured baseline. Four tautological assertions (two `|| true` disjunctions, a self-multiplying metric check, and a dead variable that was computed but never asserted) were exposed by a dedicated audit and either removed or rewritten as real checks; one of them had been masking a wrong closed-form slope in the Poincaré extension for its entire lifetime. Gate calibration is treated as a first-class activity: documented false-red episodes (a $\sqrt{2}$ scaling slip in the auditor von-Mises conversion; a Gram-Schmidt normalization threshold that silently skipped 10^{-10} -scale columns; a cross-implementation static sentinel left unsynced after a formula change) are recorded and their fixes covered by regression assertions.

5. Availability

5.1. Current code version

v7.0.0-generative-biophysics (MIT license).

5.2. Operating requirements

See Necessary resources; no backend, no telemetry, no build-time secrets.

5.3. Installation

Open `docs/platform/index.html` (static build) or run `npm install` and `npm run dev` in `tpms/tpms-platform`. CI: `npm run test:all`. A deterministic agent CLI (`tpms/agent/tpms.mjs`) exposes `list`, `estimate`, and `mesh` subcommands with JSON output for scripting and LLM-tool integration.

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